

Bhavya Khurana

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Education

Manipal Institute of Technology

B.Tech in Computer Science Engineering – Artificial Intelligence and Machine Learning

Manipal, Karnataka

2023 – 2027

CGPA: 7.93/10

Certifications: Big Data Specialization – University of California San Diego

Relevant Coursework: Deep Learning, Computer Vision, Object-Oriented Programming, Database Management Systems, Big Data Analytics

Experience

Lepton Software

2026

Data and Analytics Intern

- Built a **sub-hourly rainfall nowcasting and traffic-impact pipeline** integrating spateGAN and multi-quantile regression models to forecast road-level traffic disruption.
- Engineered a geospatial data layer by integrating data from **ERA5 reanalysis, Google Weather API and H3 spatial indexing** for operational road-impact forecasting across Pune and Kolkata.
- Backtested **59,000+ served road-hour predictions**, reducing speed error by **18–63%** versus climatological baselines.

Projects

Portfolio AI

[Live](#)

- Built an LLM-powered portfolio assistant using **FastAPI** and **OpenAI's GPT-OSS**.
- Designed a **ReAct-style agent** with custom tools for job description extraction and automated candidate-job matching.

Acquisition Robust Network

- Building an acquisition-robust **lung-nodule detection pipeline** combining physics-simulated CT augmentation, an anisotropy-conditioned 3D Swin Transformer encoder, and a cross-acquisition consistency-distillation loss
- Validated a thinning-induced detection reduction (5mm CPM reduction from **33.1% to 21.6%**) without loss of baseline detection capability,

CUDA Stereo Depth Estimation

- Developed a **CUDA-accelerated stereo depth pipeline** using Census transform, 8-path Semi-Global Matching and winner-takes-all disparity selection.
- Implemented six custom CUDA stages using **shared memory, parallel min-reduction, and GPU global-memory pipelines**, processing KITTI Stereo 2015 stereo pairs without CPU intervention.
- Achieved **7.1% Bad3.0 error, 1.78 px MAE, and 168.5 ms median processing time** across 200 KITTI scenes, delivering approximately **30× speedup** over the CPU baseline.

Adaptive Hopfield Memory with Dynamic Pattern Forgetting

- Developed **AHM-DPF**, a feedback-driven Hopfield architecture with adaptive memory updates and selective forgetting to mitigate catastrophic interference.
- Evaluated multiple Hopfield variants on a **BitPatternSet** benchmark, comparing adaptive and non-adaptive memory mechanisms under different capacity settings.
- Achieved **99.8% accuracy** in the 64-D setting and **87.3%** under fair 8-D comparison, outperforming non-adaptive variants by **25–33 percentage points**.

Network Intrusion Anomaly Detection System

[GitHub](#)

- Developed a **two-stage intrusion detection system** using supervised models for known threats and unsupervised models for zero-day anomaly detection on the UNSW-NB15 dataset.
- Achieved **99.47% accuracy** using Random Forest for known threats and Isolation Forest achieved ROC-AUC score of **0.858** for zero-day detection.

Technical Skills

Languages: Python, C++, C, Java, SQL

Libraries and Tools: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, PostGIS, PostgreSQL